Competitive Exam Practice 2: Operating-System Operations and System Protection

GATE (Computer Science) Pattern

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Concepts That Click

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern — no past-year provenance is claimed for any item.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which of the following statements about dual-mode operation is **TRUE**?

(A) The mode bit is an ordinary word in the process's data segment, so a program can set it with a normal store.

(B) Mode bit 1 denotes kernel mode and mode bit 0 denotes user mode.

(C) The hardware checks the mode bit before executing a privileged instruction, and only controlled events such as traps, interrupts, and exceptions change the mode.

(D) A user-mode program may disable interrupts, provided it re-enables them before the next timer expiry.

Q2 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which pair consists **only** of privileged instructions?

(A) Read the clock; execute a trap instruction

(B) Load the timer; change the page-table base register

(C) Add two registers; disable interrupts

(D) Branch to a subroutine; issue a raw I/O instruction

Q3 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about the trap instruction and the system-call gate is **TRUE**?

(A) The trap instruction is privileged, so only code already running in kernel mode can execute it.

(B) The trap instruction is unprivileged; it transfers control to a hardware-defined entry point, not to an address chosen by the program.

(C) After a trap, the kernel may return to any instruction address supplied by the user process.

(D) The kernel validates the system-call arguments only after it has returned to user mode.

Q4 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about memory and I/O protection is **TRUE**?

(A) Base and limit registers are compared against every address by kernel software, on each user memory access.

(B) A page table protects memory even if user code is able to write the page-table base register.

(C) Raw I/O instructions may be executed by user code because device registers are memory-mapped.

(D) An out-of-limit address traps to the kernel, and raw I/O instructions are privileged, so devices are reached only through kernel-validated system calls.

Q5 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Which of the following is a *protection* mechanism rather than a *security* policy?

(A) Requiring a password at login

(B) Checking every user address against a base-and-limit pair in hardware

(C) Installing a firewall rule that blocks a remote-access port

(D) Encrypting a backup tape before it leaves the building

Q6 *[Original, GATE-pattern]* **[NAT, 2 marks]** A context switch costs 12 μs. The scheduler uses a quantum of 250 μs. The percentage of CPU time spent on context switching is _____.

Q7 *[Original, GATE-pattern]* **[NAT, 2 marks]** A hardware timer interrupts every 1 ms. Each timer interrupt costs 25 μs to handle. The quantum is 5 ms, and one context switch costing 15 μs occurs at every quantum boundary. The total percentage of CPU time spent on timer handling and context switching is _____.

Q8 *[Original, GATE-pattern]* **[NAT, 1 mark]** A process executes 950 ms of user-mode instructions and makes 2000 system calls. Each system call spends 25 μs in kernel mode. The percentage of the process's total CPU time spent in kernel mode is _____.

Q9 *[Original, GATE-pattern]* **[NAT, 1 mark]** A runaway process executes only unprivileged instructions. The quantum is 8 ms and a context switch costs 16 μs. Over an interval of 2 s, the total time (in ms) lost to context switches forced by timer preemption is _____.

Q10 *[Original, GATE-pattern]* **[NAT, 1 mark]** A program makes 1500 system calls; each call requires two mode switches (into the kernel and back). During the same run the timer preempts the program 250 times, and each preemption likewise requires two mode switches. The total number of mode switches is _____.

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